$$\int_0^{\sqrt{2}}\left[\frac{3}{2}\cos^{-1}\left(\sqrt{\frac{2}{2+x^2}}\right)+\frac{1}{4}\sin^{-1}\left(\frac{2\sqrt{2}x}{2+x^2}\right)+\tan^{-1}\left(\frac{\sqrt{2}}{x}\right)\right]dx$$

$x := \tan t\sqrt{2}$

$dx = \sec^2 t\sqrt{2}dt$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left[\frac{3}{2}\cos^{-1}\left(\sqrt{\frac{2}{2+\left(\tan t\sqrt{2}\right)^2}}\right)+\frac{1}{4}\sin^{-1}\left(\frac{2\sqrt{2}\left(\tan t\sqrt{2}\right)}{2+\left(\tan t\sqrt{2}\right)^2}\right)+\tan^{-1}\left(\frac{\sqrt{2}}{\tan t\sqrt{2}}\right)\right]\sec^2 tdt$$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left[\frac{3}{2}t+\frac{1}{4}\sin^{-1}\left(\frac{2\tan t}{\sec^2 t}\right)+\tan^{-1}(\tan(\tfrac{\pi}{2}-t))\right]\sec^2 tdt$$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left[\frac{3}{2}t+\frac{1}{4}\sin^{-1}\left(\frac{2\sin t\cos^2 t}{\cos t}\right)+\frac{\pi}{2}-t\right]\sec^2 tdt$$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left[\frac{1}{2}t+\frac{\pi}{2}+\frac{1}{4}\sin^{-1}(2\sin t\cos t)\right]\sec^2 tdt$$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left[\frac{1}{2}t+\frac{\pi}{2}+\frac{1}{4}\sin^{-1}(\sin(2t))\right]\sec^2 tdt$$

$$\sqrt{2}\int_0^{\frac{\pi}{4}}\left(\frac{1}{2}t+\frac{\pi}{2}+\frac{1}{2}t\right)\sec^2 tdt$$

$$\sqrt{2}\left(\int_0^{\frac{\pi}{4}}t\sec^2 tdt+\frac{\pi}{2}\int_0^{\frac{\pi}{4}}\sec^2 tdt\right)$$

$u := t$ $dv := \sec^2 t$

$du = dt$ $v = \tan t$

$$\sqrt{2}\left(t\tan t\ \Big|_0^{\frac{\pi}{4}}-\int_0^{\frac{\pi}{4}}\tan tdt+\frac{\pi}{2}\tan t\ \Big|_0^{\frac{\pi}{4}}\right)$$

$$\sqrt{2}\left(\frac{\pi}{4}-\int_0^{\frac{\pi}{4}}\frac{\sin t}{\cos t}dt+\frac{\pi}{2}\right)$$

$w := \cos t$

$dw := -\sin tdt$

$$\sqrt{2}\left(\frac{3\pi}{4}+\int_1^{\frac{\sqrt{2}}{2}}\frac{1}{w}dw\right)$$

$$\sqrt{2}\left(\frac{3\pi}{4}+\ln|w|\ \Big|_1^{\frac{\sqrt{2}}{2}}\right)$$

$$\sqrt{2}\left(\frac{3\pi}{4}-\frac{1}{2}\ln 2\right)$$

$$\frac{3\pi\sqrt{2}}{4}-\frac{\sqrt{2}}{2}\ln 2$$